

Lecture 9 & 10

In Class Exercises (solutions)



Exercise 1

- Using the figure in the slides as a model, illustrate the operation of COUNTING-SORT on the array $A = \{6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2\}$.

Exercise 2

- For the set $\{1, 4, 5, 10, 16, 17, 21\}$ of keys, draw binary search trees of heights 2, 3, 4, 5 and 6.

Exercise 3

- Give a recursive algorithm that performs preorder and postorder tree walks in $\Theta(n)$ time on a tree of n nodes.

Exercise 3 (solution)

- Give a recursive algorithm that performs preorder and postorder tree walks in $\Theta(n)$ time on a tree of n nodes.

PREORDER-TREE-WALK(x)

```
if  $x \neq \text{NIL}$ 
    print  $x.\text{key}$ 
    PREORDER-TREE-WALK( $x.\text{left}$ )
    PREORDER-TREE-WALK( $x.\text{right}$ )
```

POSTORDER-TREE-WALK(x)

```
if  $x \neq \text{NIL}$ 
    POSTORDER-TREE-WALK( $x.\text{left}$ )
    POSTORDER-TREE-WALK( $x.\text{right}$ )
    print  $x.\text{key}$ 
```

Exercise 4

- Write recursive versions of TREE-MINIMUM and TREE-MAXIMUM.

Exercise 4 (solution)

- Write recursive versions of TREE-MINIMUM and TREE-MAXIMUM.

RECURSIVE-TREE-MINIMUM(x)

if $x.left == \text{NIL}$

return x

else return RECURSIVE-TREE-MINIMUM($x.left$)

RECURSIVE-TREE-MAXIMUM(x)

if $x.right == \text{NIL}$

return x

else return RECURSIVE-TREE-MAXIMUM($x.right$)

Exercise 5

- Write the TREE-PREDECESSOR procedure.

Exercise 5 (solution)

- Write the TREE-PREDECESSOR procedure.

```
TREE-PREDECESSOR( $x$ )  
  if  $x.left \neq \text{NIL}$   
    return TREE-MAXIMUM( $x.left$ )  
  else  $y = x.p$   
    while  $y \neq \text{NIL}$  and  $x == y.left$   
       $x = y$   
       $y = y.p$   
  return  $y$ 
```